

Final Project: Using SQL in MYSQL with phpMyAdmin



Estimated time needed: 20 minutes

In this lab, you will learn how to create tables and load data in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

Software Used in this Lab

In this lab, you will use [MySQL](#). MySQL is a Relational Database Management System (RDBMS) designed to efficiently store, manipulate, and retrieve data.



To complete this lab you will utilize MySQL relational database service available as part of IBM Skills Network Labs (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

Review and familiarize yourself with the datasets

To complete the assignment problems you will be using three datasets that are available on the city of Chicago's Data Portal:

1. [Socioeconomic Indicators in Chicago](#)
2. [Chicago Public Schools](#)
3. [Chicago Crime Data](#)

1. Socioeconomic Indicators in Chicago

This dataset contains a selection of six socioeconomic indicators of public health significance and a "hardship index," for each Chicago community area, for the years 2008 – 2012.

A detailed description of this dataset and the original dataset can be obtained from the Chicago Data Portal at:

<https://data.cityofchicago.org/Health-Human-Services/Census-Data-Selected-socioeconomic-indicators-in-C/kn9c-c2s2>

2. Chicago Public Schools

This dataset shows all school level performance data used to create CPS School Report Cards for the 2011-2012 school year.

A detailed description of this dataset and the original dataset can be obtained from the Chicago Data Portal at:

<https://data.cityofchicago.org/Education/Chicago-Public-Schools-Progress-Report-Cards-2011-/9xs2-f89t>

3. Chicago Crime Data

This dataset reflects reported incidents of crime (with the exception of murders where data exists for each victim) that occurred in the City of Chicago from 2001 to present, minus the most recent seven days.

A detailed description of this dataset and the original dataset can be obtained from the Chicago Data Portal at:

<https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-present/ijzp-q8t2>

This dataset is quite large - over 1.5GB in size with over 6.5 million rows. For the purposes of this assignment we will use a much smaller subset of this dataset.

Database Used in this Lab

MySQL_learners database has been used in this lab.

Here you will be creating and inserting data into the below mentioned 3 tables

- 1.chicago_public_schools
- 2.chicago_socioeconomic_data
- 3.chicago_crime

Here you will be using 3 dump files for this purpose.

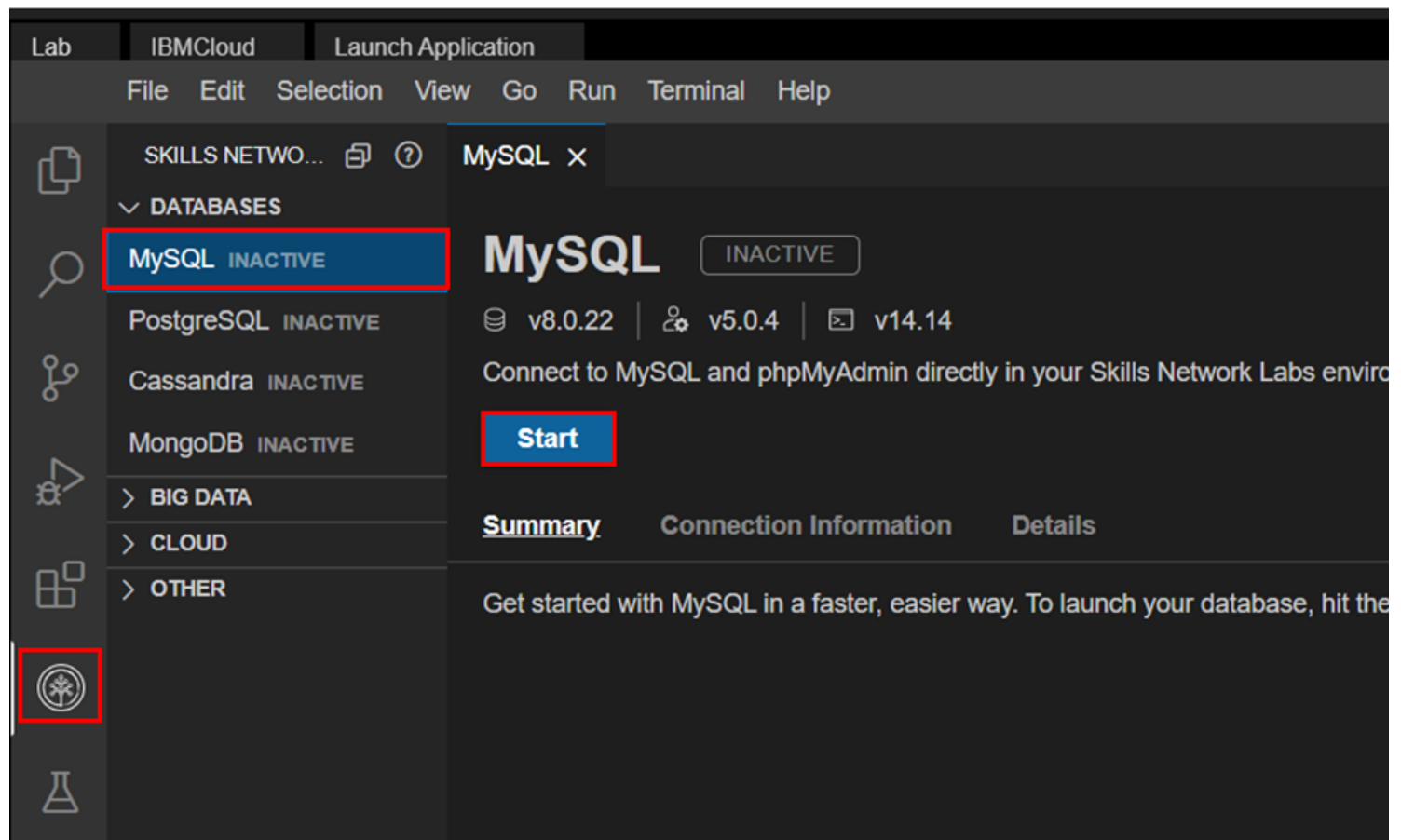
[chicago_public_schools](#)

[chicago_crime](#)

Create a database

1. Click on **Skills Network Toolbox**. In **Database** section, click **MySQL**.

To start the MySQL click **Start**.



2. Once **MySQL** has started, click on **phpMyAdmin** button to open **phpMyAdmin** in the same window.

Lab

IBMCLOUD

Launch Application

FileEditSelectionViewGoRunTerminalHelp

MySQL xphpMyAdmin

MySQL

ACTIVE

v8.0.22 | v5.0.4 | v14.14

Connect to MySQL and phpMyAdmin directly in your Skills Network Labs environment.

Stop

SummaryConnection InformationDetails

Your database and phpMyAdmin server are now ready to use and available with the following login credentials. For more details on how to navigate MySQL, please check out the Details section.

Username:malikas

Password:

You can manage MySQL via:

phpMyAdmin

Or to interact with the database in the terminal, select one of these options:

MySQL CLINew Terminal

3. You will see the phpMyAdmin GUI tool.

phpMyAdmin



Recent

Favorites

- + New
- + information_schema
- + mysql
- + performance_schema
- + sakila
- + sys

← Server: mysql:3306



Databases



SQL



Status



General settings

☰ Server connection collation: ⓘ utf8mb4_ur

🔑 [More settings](#)

Appearance settings

🗣️ Language ⓘ English

🎨 Theme: pmahomme ▼

4. In the tree-view, click **New** to create a new empty database. Then enter **Mysql_Learners** as the name of the database and click **Create**.

The encoding will be left as **utf8mb4_0900_ai_ci**. UTF-8 is the most commonly used character encoding for content or data.

Databases

SQL

Status

User accounts

Export

Import

Settings

Binary log

Replica

Databases

Create database

Mysql_learners

utf8mb4_0900_ai_ci

Create

	Database	Collation	Master replication	Action
☐	information_schema	utf8_general_ci	✔ Replicated	Check privileges
☐	mysql	utf8mb4_0900_ai_ci	✔ Replicated	Check privileges
☐	performance_schema	utf8mb4_0900_ai_ci	✔ Replicated	Check privileges
☐	sys	utf8mb4_0900_ai_ci	✔ Replicated	Check privileges

Total: 4

⬆

☐ Check all

With selected: Drop

Note: Enabling the database statistics here might cause heavy traffic between the web server and the MySQL server.

•

Enable statistics

Load the dump files one by one into the database **Mysql_learners** by clicking the **Import** tab and choose the file. Click on **Go** button.

Importing into the database "Mysql_learners"

File to import:

File may be compressed (gzip, bzip2, zip) or uncompressed.
A compressed file's name must end in **.[format].[compression]**. Example: **.sql.zip**

Browse your computer:

Choose File

chicago_pu..._schools.sql (Max: 2,048KiB)

You may also drag and drop a file on any page.

Character set of the file:

utf-8

Partial import:

☒ Allow the interruption of an import in case the script detects it is close to the PHP timeout limit. *(This might be a good way to import large files, howe*

Skip this number of queries (for SQL) starting from the first one:

0

Other options:

☒ Enable foreign key checks

Format:

Console

The screenshot displays the phpMyAdmin web interface. On the left, the 'Database: Mysql_learners' is selected in the sidebar. The main panel shows the 'Structure' tab, which lists the database's tables: 'information_schema', 'mysql', 'Mysql_learners', 'chicago_public_schools', 'performance_schema', and 'sys'. The 'chicago_public_schools' table is highlighted. The right panel shows the 'SQL' tab with a list of executed queries and their results. The first query is an import statement: 'Import has been successfully finished, 22 queries executed. (chicago_public_schools.sql)'. The subsequent queries are 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0008 seconds.)', 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)', 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)', and 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)'. The SQL statements shown are: '-- phpMyAdmin SQL Dump -- version 5.0.4 -- https://www.phpmyadmin.net/ -- -- Host: mysq', 'Version: 7.4.15 SET SQL_MODE = "NO_AUTO_VALUE_ON_ZERO"', 'START TRANSACTION', 'SET time_zone = "+00:00"', and '/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */'.

Server: mysql:3306 » Database: Mysql_learners

Structure SQL Search Query Export Import Operations

Recent Favorites

New
+ information_schema
+ mysql
- Mysql_learners
+ New
+ chicago_public_schools
+ performance_schema
+ sys

✓ Import has been successfully finished, 22 queries executed. (chicago_public_schools.sql)

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0008 seconds.)

-- phpMyAdmin SQL Dump -- version 5.0.4 -- https://www.phpmyadmin.net/ -- -- Host: mysq
Version: 7.4.15 SET SQL_MODE = "NO_AUTO_VALUE_ON_ZERO"

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

START TRANSACTION

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

SET time_zone = "+00:00"

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */

The tables are created and the data is loaded successfully. Repeat the same operation with the other 2 dump files to create and load the tables.

You will see a screen as below

The screenshot shows a MySQL IDE interface. On the left is a database schema tree with the following structure:

- Recent
- Favorites
- There are no favorite tables.
- New
- information_schema
- mysql
- Mysql_learners
 - New
 - chicago_crime
 - chicago_public_schools
 - chicago_socioeconomic_data
- performance_schema
- sys

On the right is the query editor with the following content:

```

`PERCENT_OF_HOUSING_CROWDED`, `PERCENT_HOUSEHOLDS_BELOW_POVERTY`, `PERCENT_AGED_16_UNE
`PERCENT_AGED_UNDER_18_OR_OVER_64`, `PER_CAPITA_INCOME`, `HARDSHIP_INDEX`) VALUES ('1',
Ridge', '7.8', '17.2', '8.8', '20.8', '38.5', 23040, '46'), ('3', 'Uptown', '3.8', '24.
'8.2', '13.4', '25.5', 37524, '17'), ('5', 'North Center', '0.3', '7.5', '5.2', '4.5',
60058, '5'), ('7', 'Lincoln Park', '0.8', '12.3', '5.1', '3.6', '21.5', 71551, '2'), ('

```

Below the query editor, there are four status messages indicating that MySQL returned an empty result set (i.e. zero rows) for each query. The queries are:

- MySQL returned an empty result set (i.e. zero rows). (Query took 0.0002 seconds.)
- COMMIT
- MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)
- MySQL returned an empty result set (i.e. zero rows). (Query took 0.0002 seconds.)
- MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

At the bottom of the interface is a Console tab.

Problems

Now write and execute SQL queries to solve assignment problems

Problem 1

Find the total number of crimes recorded in the CRIME table.

Problem 2

Retrieve first 10 rows from the CRIME table.

Problem 3

How many crimes involve an arrest?

Problem 4

Which unique types of crimes have been recorded at GAS STATION locations?

► [Click here for a hint](#)

Problem 5

In the CENUS_DATA table list all Community Areas whose names start with the letter 'B'.

Problem 6

Which schools in Community Areas 10 to 15 are healthy school certified?

Problem 7

What is the average school Safety Score?

Problem 8

List the top 5 Community Areas by average College Enrollment [number of students]

Problem 9

Use a sub-query to determine which Community Area has the least value for school Safety Score?

Problem 10

[Without using an explicit JOIN operator] Find the Per Capita Income of the Community Area which has a school Safety Score of 1.

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